

30 Days SQL Micro Course

Day 1 – Introduction and Installation of SQL Server

What is Data?

Data is a collection of facts(**Facts** are pieces of information that are **true and can be proven**), numbers, text, images, or information about something.

Simply put:

Data is raw information that can be stored, processed, and used to make decisions.

Examples of Data

- Name: Rahul
- Age: 22
- Salary: ₹50,000
- Temperature: 30°C
- Marks: 85

All of these are examples of data.

Types of Data

1. Numeric Data (Numbers)

- Age = 25
- Salary = 50000
- Marks = 90

2. Text Data (Words)

- Name = Priya
- City = Mumbai

3. Date Data

- 05-06-2026
- 01-01-2025

4. Boolean Data (True/False)

- Passed = True

- Active = False

Data vs Information

Data	Information
Raw facts	Processed and meaningful facts
Marks: 85, 90, 95	Average Marks = 90
Individual records	Useful result from records

Interview Definition

Data is a collection of raw facts and figures that can be processed to generate meaningful information.

Introduction to Database

A **database** is a place where information is stored in an organized way so that it can be easily found, updated, and managed.

Real-Life Example

Think of a **school notebook**:

Roll No	Name	Marks
101	Rahul	85
102	Priya	92
103	Aman	78

This table stores student information. A database works in a similar way but on a computer.

Why Do We Need a Database?

Without a database:

- Data can get lost.
- Finding information takes more time.
- Updating records is difficult.

With a database:

- Data is stored safely.
- Information can be found quickly.
- Multiple users can access data.
- Data can be updated easily.

Examples of Databases

- **Banking systems** – customer accounts and transactions.
- **Hospitals** – patient records.
- **Schools/Colleges** – student information.
- **E-commerce websites** – products, orders, and customers.
- **Social media apps** – user profiles and posts.

Brief History of database

1. Why databases were needed

- People and companies started collecting more and more data (information).
- They needed a safe and organized way to store that data.
- This led to the creation of **databases**.

2. Who started it?

- **Charles Bachman** created one of the first database systems called **IDS (Integrated Data Store)** in the 1960s.
- His work was so important that he received the **Turing Award** (the biggest award in computer science).

3. What happened next?

- In the late 1960s, **IBM** developed **IMS (Information Management System)**, another database system.
- Many organizations used it to store and manage data.

4. How SQL was born

- In the 1970s and 1980s, IBM worked on a project called **System R**.
- From this project came **SQL (Structured Query Language)**.

- SQL became the standard language for working with databases.
- Today, SQL is used to:
 - Store data
 - Retrieve data
 - Update data
 - Delete data

Database Development Process

When creating a database, people usually follow these steps:

1. Analyze

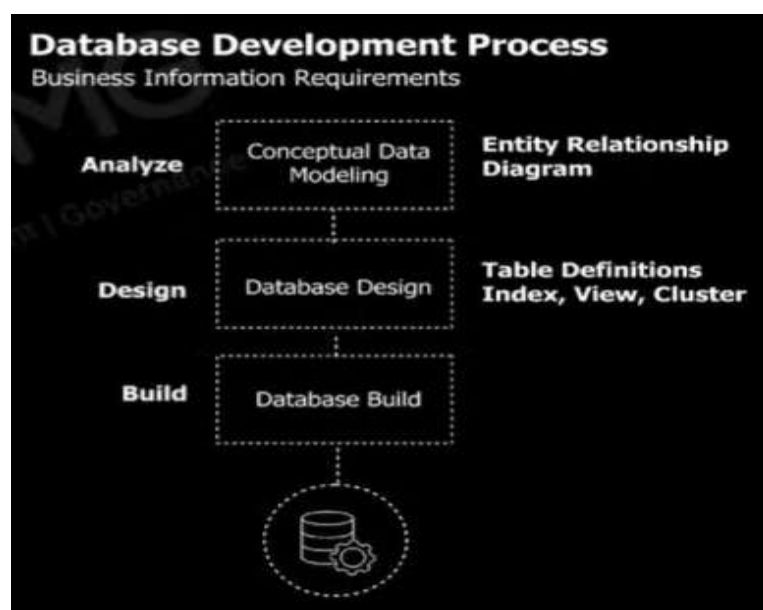
- Understand what information the business needs.
- Example: A school needs student, teacher, and course data.

2. Design

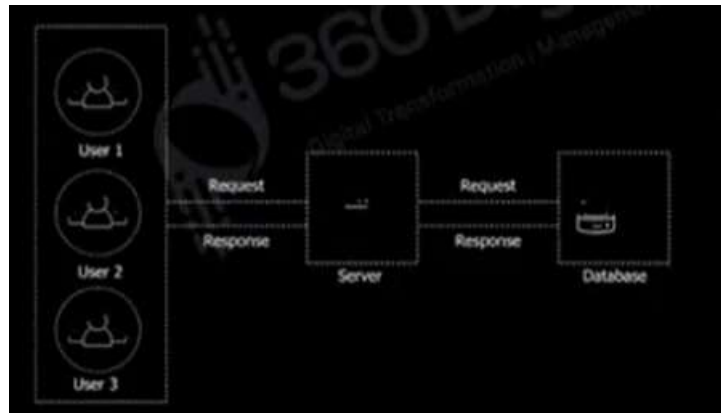
- Plan how the database will look.
- Create tables and relationships.

3. Build

- Create the actual database in software like MySQL, SQL Server, PostgreSQL, etc.



Role of the Database



This diagram shows how a **User**, **Server**, and **Database** work together.

Step 1: User Sends a Request

A user asks for information.

Example:

- "Show my bank balance."
- "Show employee details."



Step 2: Server Receives the Request

The **Server** receives the request from the user.

- The server contains application code/programs.
- It processes the user's request.
- It acts as a middleman between the user and the database.



Step 3: Server Asks the Database

The server sends a request to the database.

Example:

Get details of Employee ID = 101



Step 4: Database Finds the Data

The database stores all the data.

It searches for the requested information and sends it back to the server.



Step 5: Server Sends Response to User

The server receives the data from the database and displays it to the user.

What is a Data Type?

A **data type** tells the database **what kind of data can be stored in a column**.

Think of it like a container:

- A water bottle stores water.
- A database column stores a specific type of data.

Examples

Column Name	Example Value	Data Type	Reason
Employee_Name	Rahul	VARCHAR	Stores text (name)
Employee_ID	B56850	VARCHAR	Stores letters + numbers (alphanumeric)
Age	25	INT	Stores whole numbers
Salary	45000.50	NUMERIC / FLOAT	Stores decimal numbers
Date_of_Birth	1990-04-04	DATE	Stores only date
Login_Time	2023-03-13 17:00:00	TIMESTAMP	Stores date and time
Eligible_To_Vote	TRUE	BOOLEAN	Stores True/False values

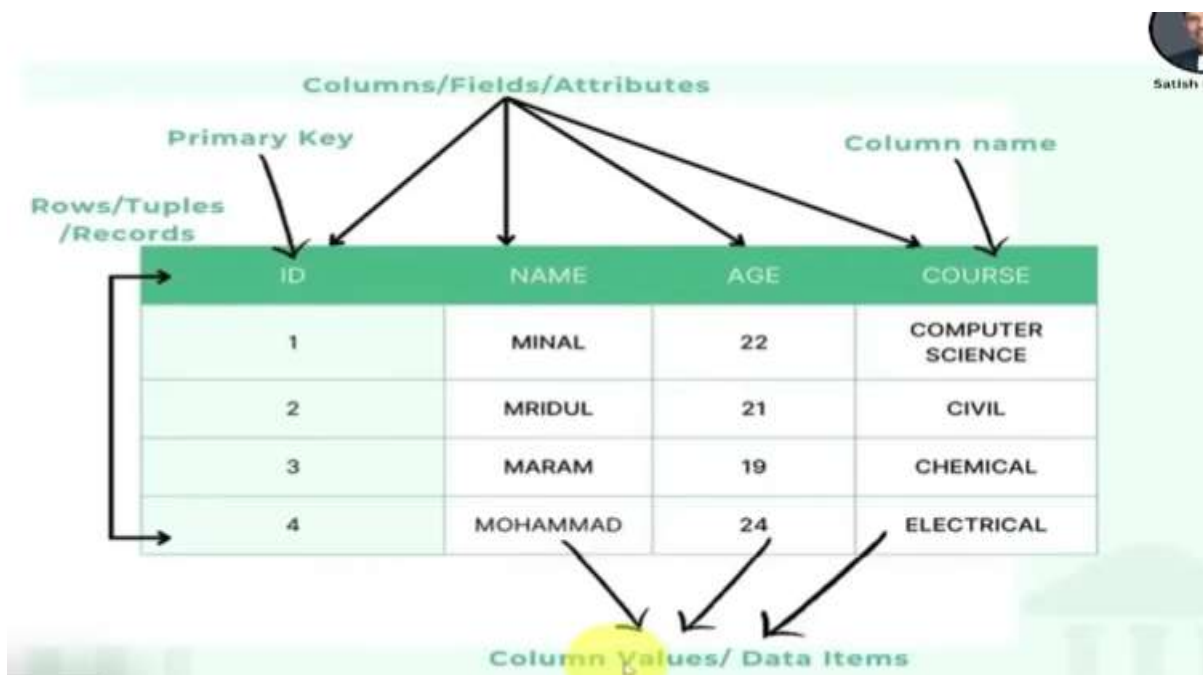
Types of Data

Type	Description	Example
Structured Data	Data organized in rows and columns	Student table, Employee table
Unstructured Data	Data without a fixed format	Images, videos, audio, emails
Semi-Structured Data	Data has some structure but not tables	JSON, XML

1. Structured Data

Data is stored in tables (rows and columns).

Example:



The diagram shows a table with the following structure:

Columns/Fields/Attributes			
ID	NAME	AGE	COURSE
1	MINAL	22	COMPUTER SCIENCE
2	MRIDUL	21	CIVIL
3	MARAM	19	CHEMICAL
4	MOHAMMAD	24	ELECTRICAL

Labels and arrows in the diagram:

- Primary Key:** Points to the ID column.
- Column name:** Points to the header row (ID, NAME, AGE, COURSE).
- Rows/Tuples/Records:** Points to the data rows.
- Column Values/ Data Items:** Points to the individual data cells (e.g., 1, MINAL, 22, COMPUTER SCIENCE).

Employee_name	Employee_id	Age	DOB	Salary	Eligibility for Perks?
A	1	30	04-04-1993	30000	Y
B	2	20		20000	N

Handwritten annotations:

- Table/Entity:** Points to the entire table structure.
- Columns/Fields/Attributes Rows/Records/Tuples:** Points to the header row.
- Handwritten:** "Table" (circled), "JPR" (signature), and a checkmark.

Important Terms

1. Table (Entity)

The entire table is called a Table or Entity.

Employee Table

Think of it as a spreadsheet storing employee information.

2. Columns (Fields / Attributes)

The headings are called Columns, Fields, or Attributes.

Employee_Name Employee_Id Age DOB Salary Eligibility for Perks

Examples:

- Employee_Name
- Employee_Id
- Age
- DOB
- Salary

Each column stores one type of information.

3. Rows (Records / Tuples)

Each horizontal line is called a Row, Record, or Tuple.

Example Row 1:

| A | 1 | 30 | 04-04-1993 | 30000 | Y |

This represents one employee's complete information.

Easy Memory Trick

 Table = Entire employee sheet

 Column = Category (Name, Age, Salary)

 Row = One employee's details

- ✓ Easy to search and analyze
 - ✓ Stored in databases like MySQL, SQL Server, PostgreSQL, Oracle, etc.
-

2. Unstructured Data

Data does not follow rows and columns.

Examples:

- Photos
- Videos
- Audio files
- PDF documents
- Social media posts

Example:

- A selfie image
- A YouTube video
- A voice recording

✗ Harder to organize in traditional databases

3. Semi-Structured Data

Data is not in tables but has some organization.

Examples:

- JSON
- XML

Example JSON:

```
{  
  "name": "Rahul",  
  "age": 25,  
  "city": "Delhi"  
}
```

Interview Answer

Data can be classified into three types: Structured Data, Unstructured Data, and Semi-Structured Data. Structured data is stored in tables, unstructured data has no fixed format, and semi-structured data has partial organization such as JSON and XML.

What is SQL?

SQL stands for Structured Query Language.

It is a language used to communicate with databases.

Simple Definition

SQL is a language used to store, retrieve, update, and delete data from a database.

Real-Life Example

Suppose you have an Employee table:

Employee_ID	Name	Salary
1	Rahul	30000
2	Priya	40000

Using SQL, you can:

1. View Data (SELECT)

```
SELECT * FROM Employee;
```

Shows all employee records.

2. Add Data (INSERT)

```
INSERT INTO Employee  
VALUES (3, 'Aman', 35000);
```

Adds a new employee.

3. Update Data (UPDATE)

```
UPDATE Employee
SET Salary = 45000
WHERE Employee_ID = 2;
```

Changes Priya's salary.

4. Delete Data (DELETE)

```
DELETE FROM Employee
WHERE Employee_ID = 3;
```

Removes an employee record.

Why is SQL Important?

- Stores data
 - Retrieves data quickly
 - Updates records
 - Deletes records
 - Manages large amounts of data
-

Where is SQL Used?

- Banks
 - Hospitals
 - Schools
 - E-commerce websites
 - Social media apps
-

Interview Definition

SQL (Structured Query Language) is a standard language used to create, access, manipulate, and manage data stored in relational databases.

Easy Memory Trick

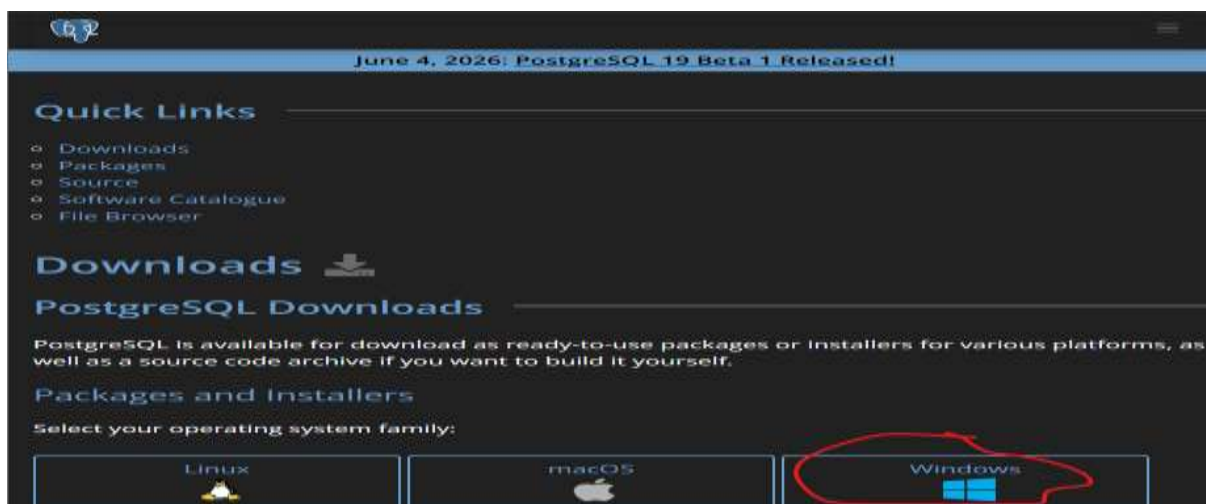
SQL = Talk to Database

SQL vs NoSQL

SQL	NoSQL
Stores data in tables (rows and columns).	Stores data in documents (JSON), key-value pairs, etc.
Looks like an Excel sheet.	Looks like a JSON file.
Has fixed columns. Every record follows the same format.	Has flexible columns. Different records can have different fields.
Best for organized and structured data.	Best for large, changing, or mixed data.
Easy to connect multiple tables using JOINS.	Usually stores related data together, so JOINS are less common.
Data accuracy and consistency are very strong.	Designed for speed and handling huge amounts of data.
Used in banks, hospitals, schools, e-commerce orders.	Used in social media, chat apps, real-time analytics.
Examples: MySQL, PostgreSQL, Oracle, SQL Server.	Examples: MongoDB, Redis, Cassandra.

Download Postgresql

Link : <https://www.postgresql.org/download/>



Quick Links

- Downloads
- Packages
- Source
- Software Catalogue
- File Browser

Windows installers

Interactive installer by EDB

Download the installer certified by EDB for all supported Postgres versions.

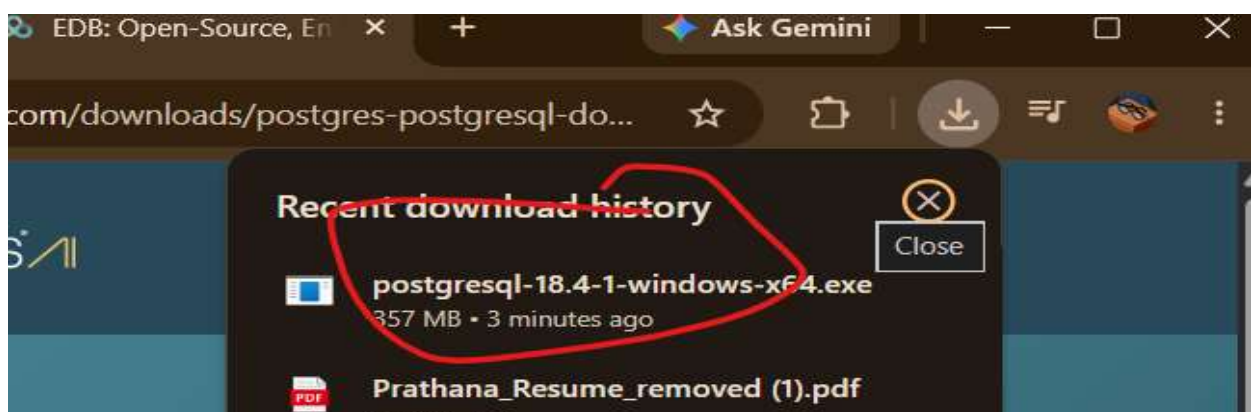
Note: This installer is hosted by EDB and not on the PostgreSQL community servers. If you have issues with the website it's hosted on, please contact webmaster@enterprisedb.com.

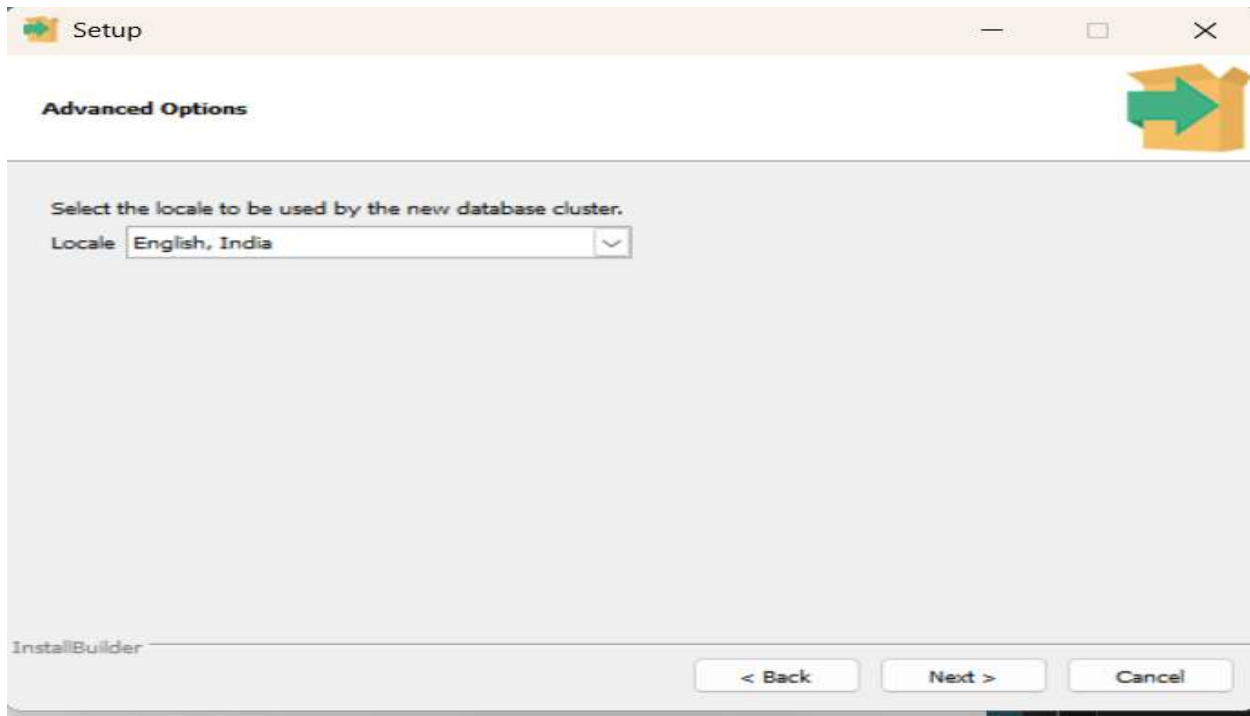
This installer includes:

Download PostgreSQL

Open source PostgreSQL packages and installers from EDB

PostgreSQL Version	Linux x86-64	Linux x86-32	Mac OS X	Windows x86-64	Windows x86-32
18.4	postgresql.org 	postgresql.org 			Not supported
17.10	postgresql.org 	postgresql.org 			Not supported





Lastly Finish .

In windows search bar search Pgadmin .

Interview que & ans

1. What is Data?

Answer:

Data is a collection of raw facts and figures that can be processed to generate meaningful information.

Example: Name, Age, Salary, Marks.

2. What is Information?

Answer:

Information is processed and meaningful data.

Example: Marks = 85, 90, 95 (Data) → Average Marks = 90 (Information)

3. What is a Database?

Answer:

A database is an organized collection of data that can be stored, managed, updated, and retrieved easily.

4. Why do we need a Database?**Answer:**

- Stores data safely
 - Easy to search data
 - Easy to update data
 - Supports multiple users
 - Prevents data loss
-

5. What is SQL?**Answer:**

SQL (Structured Query Language) is a language used to communicate with databases and perform operations like storing, retrieving, updating, and deleting data.

6. What are the main SQL commands?**Answer:**

- SELECT → View data
 - INSERT → Add data
 - UPDATE → Modify data
 - DELETE → Remove data
-

7. What is RDBMS?**Answer:**

RDBMS (Relational Database Management System) is a database system that stores data in tables and maintains relationships between those tables.

Examples: MySQL, PostgreSQL, Oracle, SQL Server.

8. What is PostgreSQL?

Answer:

PostgreSQL is a free and open-source Relational Database Management System (RDBMS).

9. What is pgAdmin?

Answer:

pgAdmin is a graphical tool used to manage PostgreSQL databases.

10. What is a Table?

Answer:

A table is a collection of related data organized in rows and columns.

11. What is a Row?

Answer:

A row (record/tuple) represents one complete entry in a table.

12. What is a Column?

Answer:

A column (field/attribute) represents a specific type of information in a table.

13. What is a Data Type?

Answer:

A data type specifies the type of data that can be stored in a column.

Examples: INT, VARCHAR, DATE, BOOLEAN.

14. Name some common SQL Data Types.

Answer:

- INT → Whole numbers
- VARCHAR → Text

- FLOAT/NUMERIC → Decimal values
 - DATE → Date values
 - BOOLEAN → True/False
-

15. What is Structured Data?

Answer:

Data organized in rows and columns is called structured data.

Example: Employee table.

16. What is Unstructured Data?

Answer:

Data without a fixed format is called unstructured data.

Examples: Images, videos, audio files.

17. What is Semi-Structured Data?

Answer:

Data that has some organization but is not stored in tables.

Examples: JSON, XML.

18. Difference between SQL and NoSQL?

SQL	NoSQL
Stores data in tables	Stores data in documents/JSON
Fixed structure	Flexible structure
Uses SQL	Uses different query methods
MySQL, PostgreSQL	MongoDB, Redis

19. What is the role of a Database in an application?

Answer:

The database stores data, the server processes requests, and the user receives the results.

Flow:

User → Server → Database → Server → User

20. What is the full form of SQL?**Answer:**

Structured Query Language.

21. What is the full form of RDBMS?**Answer:**

Relational Database Management System.

22. What is the full form of DBMS?**Answer:**

Database Management System.

23. Give examples of databases.**Answer:**

- Banking Systems
 - Hospital Management Systems
 - School Management Systems
 - E-commerce Websites
 - Social Media Applications
-

24. What is NoSQL?**Answer:**

NoSQL (Not Only SQL) is a database that stores data in flexible formats such as documents, key-value pairs, columns, or graphs.

25. What is the purpose of pgAdmin?**Answer:**

pgAdmin is used to create, manage, and query PostgreSQL databases through a graphical interface.

Most Important Interview Questions for Freshers

1. What is Data?
2. What is a Database?
3. What is SQL?
4. What is RDBMS?
5. Difference between DBMS and RDBMS.
6. Difference between SQL and NoSQL.
7. What is PostgreSQL?
8. What is pgAdmin?
9. What is a Table, Row, and Column?
10. What are Data Types?